

Kumar Gaurav

Software Engineer | Backend & Distributed Systems

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linkedin.com/in/kumar-gaurav-cs | github.com/D3S-Gaurav | Codeforces: ATHEUS | LeetCode: ATHEUS_007

Skills

Languages: Python, TypeScript, JavaScript, Go, SQL, Bash, C++ (competitive programming)

Backend: Node.js, Express.js, FastAPI, Socket.IO, WebSockets, gRPC, Kafka, REST APIs, Zod

Frontend: React, Next.js, HTML/CSS

Data & DevOps: PostgreSQL, Redis, Prisma, Docker, GitHub Actions (CI/CD), Git, Linux

Testing: k6 (load testing), Vitest, Supertest, Go testing (race detector), ESLint

Concepts: Data Structures & Algorithms, OOP, Distributed Systems, Event-Driven Architecture, Concurrency

Education

Birla Institute of Technology, Mesra | Ranchi, Jharkhand

Expected May 2027

B.Tech in Computer Science and Engineering, CGPA: 7.88/10

Experience

Backend Engineer | SoilSense (Soil Suitability Platform)

Mar 2026 – Present

Team project, Schneider Electric Hackathon 2026 | Python, FastAPI, pandas, Next.js, Gemini API | GitHub

- Wrote the crop-suitability rules engine over a 2,200-sample, 22-crop dataset, with tolerance bands (10% on NPK, 0.3 on pH) that map each deficiency to an organic amendment.
- Optimized the all-crops check to load the dataset once instead of once per crop, making the 22-crop scan **2.3× faster** (48 ms to 21 ms) and replacing 22 client requests with one.
- Hardened the Gemini analysis route against prompt injection by allow-listing and type-casting the 5 expected sensor fields, so injected keys never reach the prompt.

Open Source Contributor | Oppia, FOSSology

Jan 2026 – Present

3 merged upstream PRs | TypeScript, Angular, ESLint, CI/DevOps | Remote

- Merged a custom ESLint rule into Oppia (6.8k stars) that fails CI on undocumented `ViewEncapsulation.None`, a recurring cause of CSS leaking across components; shipped it with 10 tests and a 41-file legacy allowlist to keep CI green.
- Earned merge access to review and land PRs for Oppia's Dev Workflow team after 2 PRs passed maintainer review.
- Migrated FOSSology's Vagrant dev environment off end-of-life Ubuntu 20.04 to 22.04 to match its CI runners.

Research Intern | Birla Institute of Technology, Mesra

May 2026 – Jul 2026

NLP / Content Moderation | Python, scikit-learn, LightGBM, pandas | GitHub

- Developed a 4-class comment-moderation classifier on 198k comments, combining word/char TF-IDF with 36 engineered signals (threat and identity lexicons, intent regexes, vote-controversy ratios) in an 80k-feature matrix.
- Blended LightGBM and Logistic Regression with tuned per-class thresholds to reach **0.836 macro-F1** (5-fold CV), and raised F1 on the rare threat class (2.8% of data) from 0.23 to 0.65.

Projects

Loom | Multi-Agent LLM Orchestration Platform

Aug 2026 – Sep 2026

github.com/D3S-Gaurav/Loom | Go, TypeScript, Next.js, Kafka, gRPC, Redis, PostgreSQL, Temporal

- Architected an orchestrator that splits one goal into a dependency-aware task DAG, runs agents in parallel, and feeds validator rejections back as revisions.
- Streamed run events from Redis through Kafka and a Go service to the UI over gRPC and WebSocket, with sequence-numbered replay so reconnecting clients never apply an event twice.
- Implemented a non-blocking Go fan-out hub that drops slow subscribers instead of stalling the Kafka consumer, verified with race-detector tests.

ChatFlow | Real-Time Group Chat

Jun 2026 – Jul 2026

github.com/D3S-Gaurav/chat-fullstack | TypeScript, Node.js, Express, Socket.IO, Prisma, PostgreSQL, k6

- Load-tested with k6 to **5,000 concurrent sockets** and **~97k message deliveries/s with zero loss** on one Node process (p95 40 ms at 1,000 sockets); traced the ceiling to the single-threaded broadcast, not connection count.
- Designed the real-time backend around a typed Socket.IO event contract (TypeScript generics), so every emit/on is compile-time checked, with Zod validation on every REST and socket payload.
- Secured it with scrypt hashing, timing-safe comparison, JWT socket auth, tiered rate limiting and role-based access; covered by 32 integration tests in GitHub Actions against PostgreSQL 17.

Achievements

- Ranked 682 of 17,000+ participants (top 4%) in Codeforces Round 1102 (Div. 2); max rating 1306.
- Solved 600+ DSA problems across LeetCode, Codeforces, CodeChef and GeeksforGeeks.
- Finalist, SEBI Hackathon, a national-level hackathon hosted by the Securities and Exchange Board of India.
- Achieved Gold Level on WorldQuant BRAIN with 10,000+ points across 8 peer-reviewed alphas.